

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH



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Assignment Title: Smart Agricultural manufacturing (Cultivation and preservation) by weather analysis using IoT Software with integrated AI.

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Smart Agricultural manufacturing (Cultivation and preservation) by weather analysis using IoT Software with integrated AI.

State of knowledge:

We know that Bangladesh is a developing country, and the people of Bangladesh are mostly depended on Agriculture. We know that the unemployment problem is growing day by day and this project aims to develop an innovative smart agriculture software platform designed to enhance the cultivation, preservation, and market analysis processes for farmers. With this system we will motivate or more interact with the user who is interested in agriculture. From a beginner level user, they will get knowledge about agriculture. In fact, The Agro doctors will benefit from this system.

The system is very easy to connect with the user to guide the next generation to cultivate food. The system will analyze the weather to get the proper instructions about food cultivation. The system integrated by artificial intelligence (AI). The AI will provide both (preservation and cultivation) guidelines which can cultivate food easily and decrease the unemployment problem in future from this system, any kind of user can utilize and reduce time wastage. The AI will also analyze the market price predictions. The system will provide a online service feedback for more information about agriculture. Several Agro doctors will register and give treatment for the Infected crops.

From this system, there will be job opportunities for the future generation. The users will provide more food which will solve our nourishment problem. This system will benefit developing countries which are lag behind from other developed countries.

The key features of this system are AI driven instructions to inform and best cultivation practice, analyzing by AI model previous and current market prices for future trends, Agro experts register to provide guidelines and treatment solutions for crop diseases, alerting the farmers when to sell their produce for maximum profit, online mechanism for user's queries, additional information, preservation techniques for reduce wastage and maintain quality.

Problem of Statement:

Agriculture is a big concern for Bangladesh's economy, livelihood, and a specific portion of the population. Bangladesh's agriculture faced some challenges especially efficiency, productivity and sustainability because of having limited modern tools and technology. Farmers are unaware of the weather or struggle to predict, unaware of preservation and cultivation practices, lack of market price. These problems show the result of crop loss, resource wastages, economic instability, growing unemployment and food crisis. There is a lack of guidance of farming, lack of connection between farmers and agro experts, lack of knowledge about diseases or untreated land which are one of the reasons of reducing yields.

This research paper aims to the challenges by developing a smart agricultural manufacturing system that uses IoT software and AI integration. The system will provide weather-based cultivation, preservation and guidance to reduce food wastage and also predict the market price by analyzing previous price predictions. The best infrastructure is the AI driven decision-making and expert guidance feature, farmers more empower about farming, reduce wastage and grow more economical profits. From this system, there will create more job opportunities, food availability and more sustainable agriculture practices, especially developing countries like Bangladesh. This innovative approach will make it easy for the next generation of farmers, giving a solution to improve agricultural efficiency and socio-economic challenges by rural communities.

Primary Research Question:

How will smart agricultural manufacturing system improve cultivation efficiency, preservation practices and market analysis to improve productivity and economic stability for farmers by using IoT software with AI integrated in developing countries like Bangladesh?

Objectives:

1. To explore how IoT validated Weather analysis can optimize cultivation practices by giving real time, AI data driven recommendations to environmental conditions.
2. To analyze the advantages of AI driven models in predicting market price trends and delivering preservation guidelines to reduce food wastage.

Hypothesis:

Using IoT software for weather analysis and AI driven decision making into an agricultural manufacturing system will improve cultivation efficiency, reduce food wastage, optimize preservation techniques and improve economic profitable for farmers by providing market price predictions. The farmers will get the agro experts' consultation for any kind of inconvenience. The unemployment problem will reduce and encourage the next generation in agriculture.

Literature Review:

Smart Agriculture is smart farming which uses technology IoT (Internet of Things) and AI (Artificial Intelligence) to improve efficiency, productivity and sustainability. As global food demand increases and climate change becomes unpredictable, make stability in economic. This literature review surveys recent research and studies that mention the potential of IoT, applications and challenges of using IoT and AI in agricultural systems.

IoT in Agriculture:

An automated agriculture system is developed to monitor and maintain the important aspects of farming like temperature, humidity, soil moisture content and sunlight using IoT technology [1]. IoT uses sensors, smart devices, cloud storage, and the internet to exchange real-time data. The author also shows the importance of scalability like in developing countries struggle with the high cost of IoT devices.

IoT sensors linked with cloud-based systems might automate irrigation schedules, decreasing manual involvement while assuring appropriate water distribution.[2]. In developing countries, a very concerning factor, where water management is critical for high yield agricultural.

AI in Cultivation and Preservation:

AI have the ability to process a large number of datasets and provide in a easiest way to get the actionable component of modern agriculture. The AI model will help to analyze the weather demands. The AI driven decision-making system will instruct the user for the upcoming and ongoing problems can face while preserving the crops. AI models trained on historical weather and crop yield data can forecast optimal planting and harvesting times [3]. The AI will be the

solution to reducing post-harvest losses. The AI will maintain food quality and address wastage, especially in developing countries.

Market Price predictions:

Market analysis is critical for farmers to get the maximum profits. For market price predictions, the AI model is used for the trends. The AI model predicted the crop prices based on historical data, weather analysis and market trends. The AI model will predict the probability of trends for upcoming trends.

Challenges in implementation:

To implement the IoT technology and AI, there are some challenges to adapt and identify the issues. The challenges: high costs of technology, lack of technical expertise among farmers, and inadequate infrastructure in rural areas. The authors called for government subsidies, training programs, and the development of affordable, user-friendly technologies to bridge these gaps [4].

The IoT devices generate large amounts of sensitive data, ensure secure storage and transmission is critical to prevent misuse.

Conclusion:

The reviewed literature highlights the revolutionary power of IoT and AI in tackling crucial agricultural issues. IoT technology gives real-time environmental data for accurate decision-making, and AI-powered models improve cultivation, preservation, and market research. Despite the obvious benefits, issues like cost, accessibility, and infrastructure persist. Addressing these difficulties through targeted interventions and policy assistance has the potential to lead to broad adoption, particularly in developing countries such as Bangladesh. This analysis lays a solid platform for future research into developing a comprehensive smart agricultural manufacturing system adapted to the demands of small-scale farmers in resource-constrained regions.

Selection of Design:

This study uses a mixed-methods design, combining both quantitative and qualitative methods to identify the challenges and opportunities in smart agriculture manufacturing by IoT and AI integration.

Reason for selection:

1. Quantitative Design:

This method includes some collection of structural data based on variables such as weather analysis, yields outcome, market trends and user feedback on the system's efficiency. It provides evidence of systems' impact on agriculture, validating the hypothesis with statistical analysis.

2. Qualitative Design:

This method focuses on gathering perception from farmers and agro-doctors about their experiences, needs and expectations. It explores unstructural data such as user challenges to adapting smart technologies or the socio-economic complications of the system.

3. Mixed-methods Design:

The combination of methods will analyze the gap between numerical data and subjective user experiences. It improves the problem finding and understanding of both technical and human aspects of smart agriculture.

Research Variables:

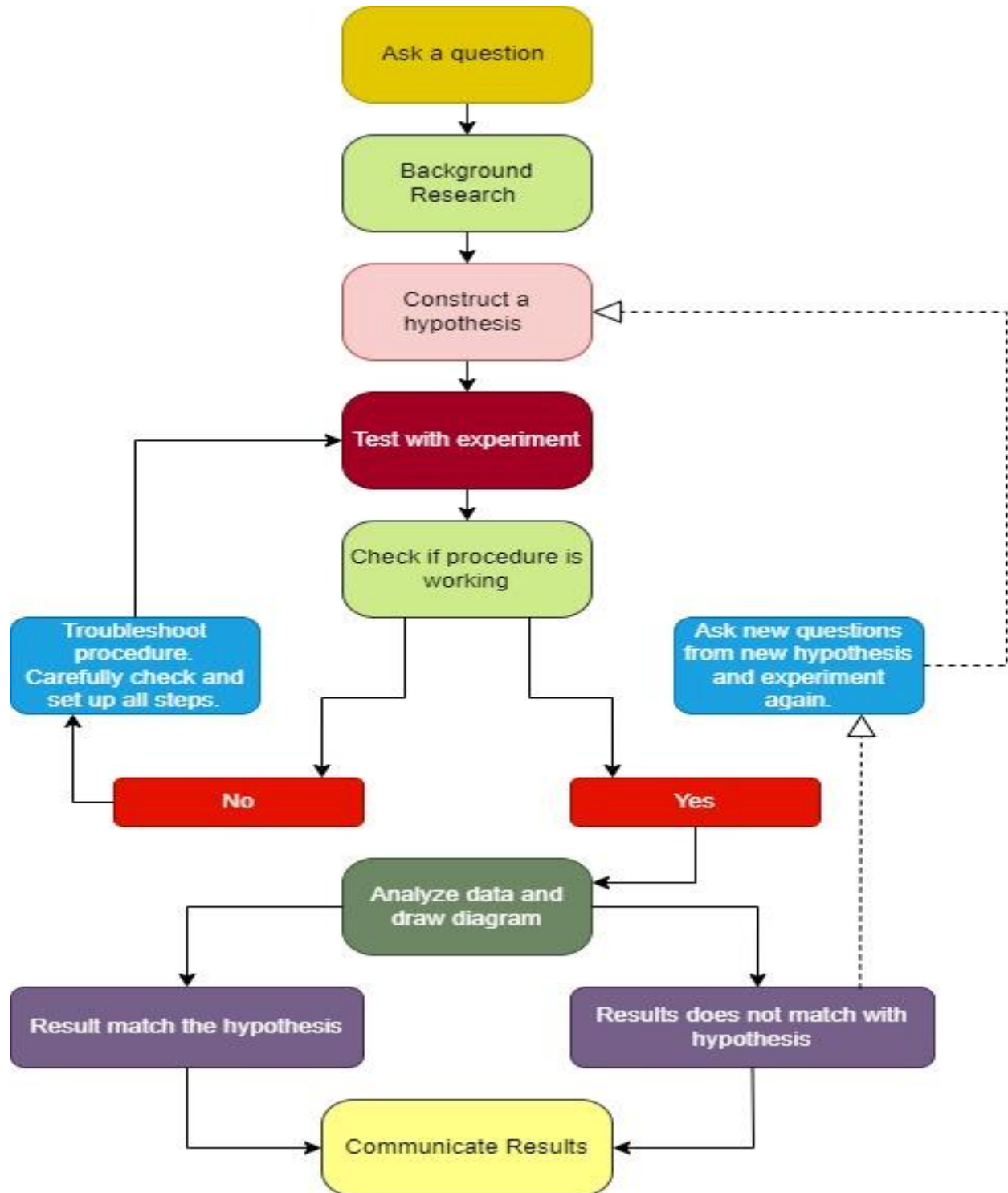
1. Independent Variable:

- **Weather Analysis Using IoT Software with Integrated AI:** This variable represents the technology and method used to analyze weather patterns and provides guidelines for agricultural practices.

2. Dependent Variable:

- **Enhancement in Cultivation, Preservation, and Market Analysis Processes for Farmers:** These variable measures the outcomes and improvements in agricultural processes resulting from the implementation of the IoT and AI-based weather analysis system.

Scientific Method Design:



References:

1. Deepa, B., Anusha, C., Chaya Devi, P. (2021). Smart Agriculture Using IOT. In: Satapathy, S., Bhateja, V., Janakiramaiah, B., Chen, YW. (eds) Intelligent System Design. Advances in Intelligent Systems and Computing, vol 1171. Springer, Singapore.
2. Alam, M., Hossain, F., & Rahman, S. (2023). Smart irrigation systems using IoT: A sustainable approach. *International Journal of Agricultural Research*, 19(4), 45-56.
3. Sharma, R., & Gupta, N. (2023). AI applications in modern farming: A case study on crop yield optimization. *Journal of Agricultural Informatics*, 12(2), 90-103.
4. Patel, R., Kumar, S., & Singh, P. (2022). Overcoming barriers to smart farming adoption. *Sustainable Agricultural Practices*, 15(1), 98-110.